Leverage Procyclicality in Chinese Financial Institutions

(Groupwork)

Task 1: Data Processing and Descriptive Statistics

1. Introduction

This report outlines the data processing methodology and descriptive statistics used to analyze leverage procyclicality among Chinese financial institutions. The analytical approach closely follows Cincinelli et al. (2021), and the dataset spans from Q4 2005 through Q4 2024.

2. Data Source and Structure

Data Source

The dataset contains quarterly financial information for publicly listed Chinese financial institutions.

Raw Data Structure

- **Format:** Wide format organized into three sheets (Banks, Finance Services, Real Estate Finance)
- **Variables per firm:** Price, Total Assets, Market-to-Book Value, Market Value, Leverage
- **Coverage period:** Q4 2005 through Q4 2024 (77 quarters in total)
- **Sample size:** 158 financial institutions

Panel Data Transformation

The original wide-format data was restructured into long-format panel data, where each row corresponds to a single firm-quarter observation. The final dataset comprises 12,166 firm-quarter observations.

Category	Number of Firms	Observations
Commercial Banks	13	1,001
Finance Services	16	1,232
Real Estate Finance	129	9,933
Total	158	12,166

3. Variable Definition

Variable	Definition	Formula
total_assets	Total Assets	Raw data (millions CNY)
leverage	Accounting Leverage	Total Assets / Common Equity
mktvalue	Market Capitalization	Share Price × Shares Outstanding
mtbv	Market-to-Book Value	Market Value / Book Value
ln_total_assets	Log Size	ln(Total Assets)
ln_leverage	Log Leverage	ln(Leverage)
ln_mktvalue	Log Market Value	ln(Market Value)
Assets_Growth	Size Growth	$\Delta \ln(TA) = \ln(TA_t) - \ln(TA_{t-1})$
Leverage_Growth	Leverage Growth	$\Delta \ln(Lev) = \ln(Lev_t) - \ln(Lev_{t-1})$
MV_Growth	Market Value Growth	$\Delta \ln(MV) = \ln(MV_t) - \ln(MV_{t-1})$
ln_levlag	Lagged Log Leverage	$\ln(\text{Leverage})_{t-1}$
mtbvlag	Lagged Market-to-Book	$MTBV_{t-1}$

4. Data Cleaning Methodology

Missing Value Treatment

Missing observations were handled based on their position within each entity's time series:

- **Internal Gaps:** Missing values surrounded by valid data points were filled through linear interpolation.
- **Leading NAs:** Missing values occurring before the first valid observation were preserved as NA to avoid artificially reconstructing historical data.
- **Trailing NAs:** Missing values appearing after the last valid observation were kept as NA to prevent fabrication of future values.

Winsorization

Rather than applying conventional percentile trimming (such as capping at the 1st and 99th percentiles), a threshold based on three times the Interquartile Range was employed. This approach targets only the most extreme statistical outliers (affecting less than 0.2% of observations) while preserving the fat-tailed distributions that are characteristic of large state-owned banks or periods of heightened market volatility.

Lagged Variables

Lagged variables were constructed for each firm individually using the shift operation. As expected, this results in a missing value for each firm's initial observation (158 firms × 1 period = 158 missing values), which is the correct and anticipated outcome of this procedure.

5. Summary Statistics

Variable	Mean	Std. Dev.	Min	Max
Total Assets (thousand CNY)	99,543,247	485,362,285	18	6,878,086,705
Size (ln)	14.417	2.604	2.890	22.652
Size growth (%)	2.5	10.8	-31.6	55.7
Market Leverage (×)	4.903	41.676	-509.540	3,285.420
Market Leverage Growth (%)	2.0	25.3	-65.7	87.2
Accounting Leverage (×)	5.342	6.042	-5.415	36.754
Accounting Leverage Growth (%)	0.7	9.5	-25.2	26.8

Table — Chinese Financial System (full sample). Note: the two leverage-level rows are ratios (×), not percentages — corrected from the original draft, which mislabelled them "(%)"; the growth rows were already correctly expressed as percentages.

Category	Total Assets (mean, th. CNY)	Size (ln, mean)	Accounting Leverage (×, mean)
Commercial Banks	1,131,318,910	20.221	16.294
Finance Services	17,740,848	15.065	3.678
Real Estate Finance	5,711,889	13.752	4.444

Table — Summary statistics by entity category

6. Stylized Facts and Notable Observations

6.1 Size

Commercial banks are by far the largest components of the sample. On average, a commercial bank holds about 1.13 trillion CNY in total assets, whereas finance services firms and real estate developers hold on average, respectively, around 17.7 billion CNY and 5.7 billion CNY. There is also a huge size gap: banks are roughly sixty times larger than finance services and nearly two-hundred times larger than real estate firms. Banks also show very stable asset growth with the lowest standard deviation (3.8%), which makes sense given their state-backed nature.

6.2 Accounting Leverage

There is a clear difference in how these institutions fund themselves. Commercial banks run at about 16× leverage on average, which is typical for deposit-taking banks operating under fractional reserve systems. Finance services and real estate firms are much less leveraged, at around 3.7× and 4.4× respectively. One interesting finding is that banks actually show slightly negative leverage growth (−0.3%), suggesting they have been slowly deleveraging over the sample period — probably due to tighter regulations after 2008.

6.3 Market Leverage

Market leverage tells a completely different story. Finance services firms have an extremely high average (11.3) but with massive volatility — the standard deviation is 110.9, almost 10 times the mean, suggesting wild swings in market valuations for this sector. Banks, on the other hand, have very low and stable market leverage (1.4 on average). Real estate firms sit somewhere in the middle at 4.5.

6.4 Growth

Looking at growth rates, finance services firms grow the fastest in terms of both assets (3.3% quarterly) and market value (3.1%). Banks grow more slowly but steadily at around 3% per quarter. Real estate firms have the slowest growth (2.3%) but the highest variability.

6.5 What This Means for Systemic Risk

- **Banks:** classic "too big to fail" institutions — enormous, highly leveraged, but operationally stable because the government stands behind them.
- **Real Estate Finance:** probably the most worrying group from a stability perspective. They are volatile across all metrics and have been increasing leverage over time; given how important real estate is to the Chinese economy, problems here could spread quickly.
- **Finance Services:** very sensitive to market sentiment. The extreme volatility in market leverage shows how quickly valuations can swing in this sector, which could amplify shocks during stress periods.

Leverage Procyclicality Analysis

1. Methodology

The methodology used to analyse the relationship between balance sheet size and financial leverage is the same as that used by Adrian & Shin (2010) and Cincinelli et al. (2021). Specifically, the leverage gap is calculated using the following formula:

$$\Delta Leverage_{i,t} = \alpha + \beta_1 \, ln(Leverage_{i,t-1}) + \beta_2 \, \Delta Size_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t}$$

- $\Delta Leverage_{i,t}$: Quarterly change in leverage (Leverage_Growth)
- $ln(Leverage_{i,t-1})$: Natural logarithm of lagged leverage
- $\Delta Size_{i,t}$: Quarterly change in total assets (Assets_Growth)
- μi: Entity fixed effects — τt: Time fixed effects

An extended model incorporates market-to-book value to account for market valuation effects, adding β_3 ln(MTBV$_{i,t-1}$) to the equation above.

2. Empirical Results

Variable	Model 1 (Baseline)	Model 2 (Extended)
Assets_Growth (β_2)	0.4420*** (20.24)	0.4465*** (21.19)
ln_levlag (β_1)	-0.0131*** (-6.01)	-0.0163*** (-6.63)
ln_mtbvlag (β_3)	–	0.0158*** (5.32)
Entity / Time Fixed Effects	Yes / Yes	Yes / Yes
Observations	11,592	11,592
R-squared	0.2395	0.2497

*Table 1. Panel regression results — full sample (2005:Q4–2024:Q4). T-statistics in parentheses; standard errors clustered at entity level. *** p<0.01, ** p<0.05, * p<0.10.*

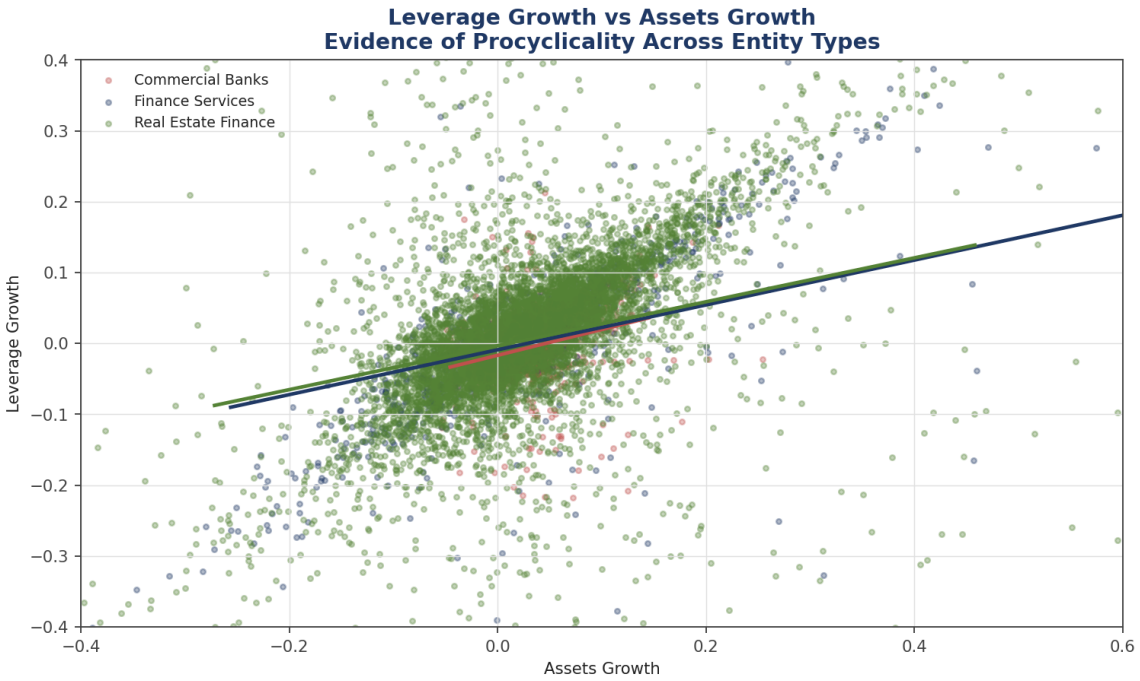


Figure 1. Leverage growth vs. assets growth, evidence of procyclicality across entity types

3. Interpretation of Results

3.1 Evidence of Procyclical Leverage ($\beta_2 = 0.4465^{***}$)

The coefficient relating to asset growth (Assets_Growth) is positive and, based on t-value tests, also highly significant. These results provide solid evidence that financial leverage in the Chinese financial market exhibits procyclical behaviour. From the linear regression, it can be seen that a 1% increase in total assets corresponds to an increase of approximately 0.45 percentage points in the growth of financial leverage. This result is consistent with the theoretical framework proposed by Adrian & Shin (2010), who contend that financial institutions actively manage their balance sheets in response to fluctuations in asset values. During periods of economic expansion, rising asset prices increase the value of collateral, allowing institutions to take on additional debt and expand their balance sheets — a dynamic that creates a virtuous circle between asset values, borrowing capacity and further asset acquisitions.

3.2 Mean Reversion in Leverage ($\beta_1 = -0.0163^{***}$)

The coefficient for delayed leverage is negative and significant ($\beta_1 = -0.0163$, $t = -6.63$), indicating that Chinese financial institutions exhibit mean-reverting behaviour in their leverage management: institutions tend to reduce leverage in subsequent periods when it exceeds the target level, and vice versa. The magnitude of -0.0163 implies a relatively gradual adjustment process — approximately 1.6% of any deviation from the target is corrected each quarter, meaning that full adjustment takes a long time.

3.3 Market Valuation Effects ($\beta_3 = 0.0158^{***}$)

The coefficient on the delayed price/book ratio is positive and significant ($\beta_3 = 0.0158$, $t = 5.32$). This indicates that institutions with a higher market valuation than book value tend to increase their leverage more markedly — consistent with Market Timing theory, whereby high market valuations lower the cost of equity financing and encourage institutions to replace debt with equity, thereby increasing financial leverage.

4. Comparison with Cincinelli et al. (2021)

Our findings are qualitatively consistent with those of Cincinelli et al. (2021), who examine leverage procyclicality in Chinese financial institutions over the 2006–2019 period. Both analyses confirm the presence of procyclical leverage behavior.

Variable	Our Study (2005–2024)	Cincinelli et al. (2006–2019)
Assets_Growth	0.4465***	0.1815***
ln_levlag	-0.0163***	-0.1685***
R^2	0.2497	0.09
Observations	11,592	10,162

The quantitative differences likely stem from several factors: (1) our sample comprises 158 financial institutions, whereas Cincinelli et al. analyze 264 institutions with a larger share of Real Estate Finance Developers; (2) our dataset extends through the COVID-19 pandemic period (2020–2024), which introduced extraordinary market volatility and government interventions; and (3) variations in data providers may lead to divergent variable calculations, particularly for leverage ratios. Despite these differences, both studies arrive at the same fundamental conclusion: leverage in Chinese financial institutions is procyclical, which carries important implications for financial stability and macroprudential policy.

5. Conclusions

This analysis provides robust evidence of leverage procyclicality in Chinese financial institutions over the 2005–2024 period. The results demonstrate that leverage is significantly procyclical ($\beta = 0.4465$, $p < 0.01$), institutions exhibit mean-reverting behavior toward target leverage levels, and market valuations positively influence leverage dynamics. These findings accord with the theoretical predictions of Adrian & Shin (2010) and the empirical results of Cincinelli et al. (2021), confirming that procyclical leverage remains a persistent feature of the Chinese financial system that warrants ongoing regulatory attention.

Panel Analysis with Marginal Effects for Financial Entities

1. Introduction and Methodology

The final segment of our analysis investigates whether leverage procyclicality differs across financial entity types and crisis periods. Always following the paper published by Cincinelli et al. (2021), we identified four sub-periods that capture distinct phases of financial conditions in China:

Period	Date Range	Characterization
GFC	2007:Q1 – 2009:Q4	Global Financial Crisis
MPR	2010:Q1 – 2014:Q4	Monetary Policy Restriction by PBoC
PMPR	2015:Q1 – 2019:Q4	Post-MPR & Chinese Stock Market Crash
COVID	2020:Q1 – 2023:Q4	COVID-19 Pandemic (our extension)

The formula used to calculate the leverage gap with interaction terms adds two interaction terms to the baseline model:

$$\Delta Leverage_{i,t} = \alpha + \beta_1(Assets_Growth) + \beta_2(AG \times FS) + \beta_3(AG \times REF) + \beta_4\, ln(Lev_{t-1}) + \beta_5\, ln(MTBV_{t-1}) + \mu_i + \tau_t + \varepsilon_{i,t}$$

Where Commercial Banks serve as the baseline group, and the interaction terms capture differential effects for Finance Services (FS) and Real Estate Finance (REF).

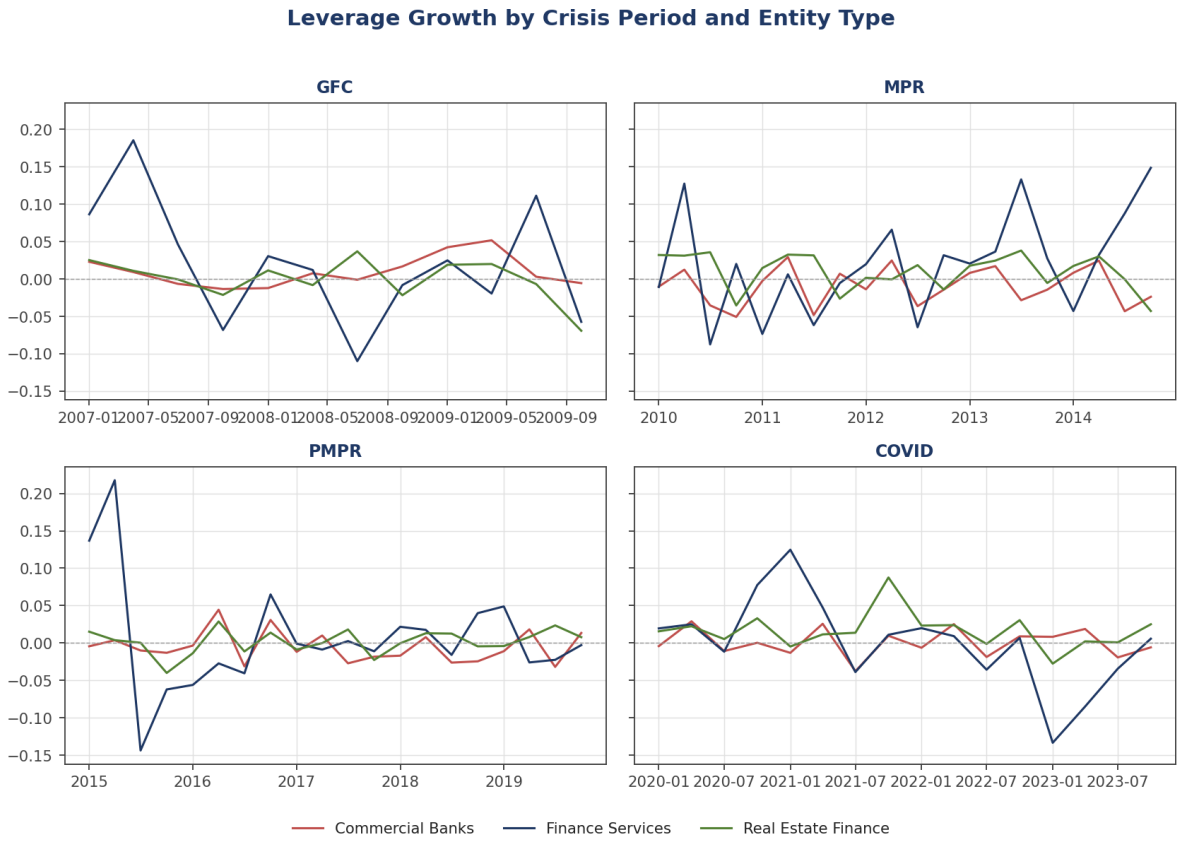


Figure 2. Leverage growth by crisis period and entity type

2. Results by Entity Type (Full Sample)

Variable	Commercial Banks	Finance Services	Real Estate Finance
Assets_Growth (β)	0.5555*** (8.37)	0.4447*** (10.28)	0.4414*** (18.78)
ln_levlag	-0.0812*** (-6.83)	-0.0393*** (-6.57)	-0.0137*** (-5.61)
ln_mtbvlag	0.0143 (1.50)	0.0358*** (8.68)	0.0132*** (4.25)
R^2	0.2109	0.3404	0.2320
Observations	988	1,204	9,400

*Table 2. Leverage procyclicality by entity type (2005:Q4–2024:Q4). T-statistics in parentheses. *** p<0.01, ** p<0.05, * p<0.10.*

The results clearly show that all variables considered are highly significant, with a p-value lower than 0.01. Commercial banks show the strongest procyclicality (β = 0.556); the other two types of financial institutions show procyclicality levels approximately 20% lower. All entity types display significant mean reversion (negative ln_levlag coefficients), especially commercial banks (−0.081). The negative interaction terms indicate they respond less aggressively than banks.

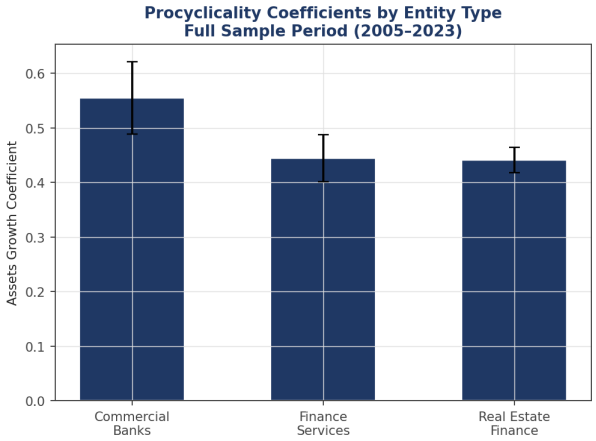


Figure 3. Procyclicality coefficients by entity type, full sample period

3. Results by Sub-Period (All Entities)

Variable	GFC (2007-09)	MPR (2010-14)	PMPR (2015-19)	COVID (2020-23)
Assets_Growth (β)	0.3141*** (9.07)	0.6304*** (16.27)	0.4467*** (11.11)	0.5203*** (11.20)
ln_levlag	-0.0543*** (-6.44)	-0.0233*** (-3.23)	-0.0565*** (-5.59)	-0.0329*** (-3.87)
ln_mtbvlag	0.0388*** (3.76)	0.0305*** (3.87)	0.0407*** (5.38)	0.0238*** (4.03)
R^2	0.2297	0.4326	0.2840	0.2483
Observations	1,766	3,097	3,129	3,025

Table 3. Leverage procyclicality by crisis period, all financial institutions. T-statistics in parentheses.

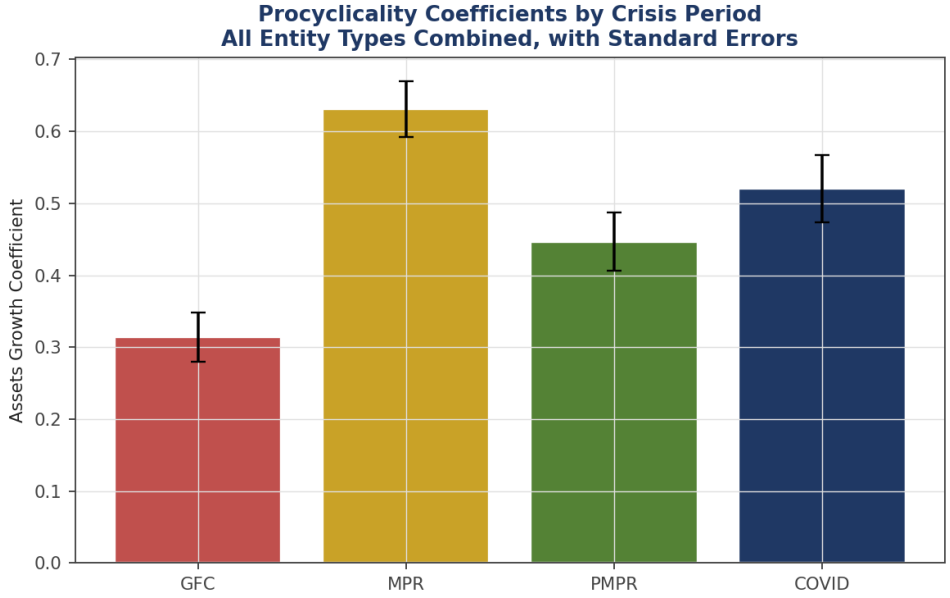


Figure 4. Procyclicality coefficients by crisis period, all entity types combined, with 95% confidence intervals

Key Findings

The MPR period exhibits the highest level of procyclicality (β = 0.63), which is consistent with the expansion of the shadow banking system that characterized those years. In contrast, the GFC period displays the lowest degree of procyclicality (β = 0.31), reflecting the heightened caution and risk aversion induced by the financial crisis. During the COVID period, procyclicality remains relatively high (β = 0.52), largely driven by government stimulus measures that sustained lending activity despite the economic downturn. Across all periods, there is evidence of mean reversion, as indicated by the negative coefficient of the lagged leverage term, suggesting a tendency for leverage levels to adjust back toward equilibrium over time.

4. Interaction Effects: Entity Type × Crisis Period

Period	CB Effect	FS Effect	REF Effect	R^2	N
GFC	0.6396***	0.4138	0.2869*	0.2343	1,766
MPR	0.7833***	0.4516	0.6845	0.4429	3,097
PMPR	0.7074***	0.5261**	0.4272***	0.2863	3,129
COVID	0.4564***	0.6263	0.5060	0.2498	3,025
Full Period	0.5281***	0.4714	0.4402	0.2500	11,592

Table 4. Procyclicality by period and entity type (interaction model). CB Effect is the base coefficient; FS and REF effects are total effects (base + interaction); significance levels based on interaction-term t-statistics.

Period	Assets_Growth (CB)	AG × FS	AG × REF
GFC	0.6396*** (3.49)	-0.2258 (-1.10)	-0.3527* (-1.88)
MPR	0.7833*** (5.38)	-0.3316** (-1.96)	-0.0987 (-0.65)
PMPR	0.7074*** (11.70)	-0.1814** (-2.05)	-0.2802*** (-4.16)
COVID	0.4564*** (5.32)	+0.1699 (1.22)	+0.0496 (0.65)

Table 5. Interaction coefficients by period. T-statistics in parentheses; negative interaction = less procyclical than CB.

5. Interpretation of Results

During the global financial crisis (2007–2009), the commercial banking (CB) system showed the highest procyclicality with a coefficient of 0.64, while the real estate sector (REF) recorded the lowest value at 0.29. The model clearly reflects the central role of banks in the distribution of fiscal and monetary stimuli, while caution in the real estate sector stems from the vulnerability of the market after the crisis. These results are consistent with those reported by Cincinelli et al. (2021), who highlight strong banking procyclicality during this critical period.

During the period of monetary policy tightening (2010–2014), the highest overall procyclicality of the entire sample is observed. The CB coefficient reaches 0.78, a value very close to the 0.82 documented by Cincinelli et al. (2021). At the same time, the REF also shows high values (probably derived from the real estate boom) at 0.68.

The post-MPR period and the stock market crash (2015–2019) have particular characteristics: the CB maintains high procyclicality (0.71), but significant negative interactions emerge for both financial services (−0.18**) and real estate (−0.28***).

The COVID-19 pandemic (2020–2023) marks a notable reversal from previous periods. Financial services (FS) become the most procyclical sector with a coefficient of 0.63, while commercial banks record the lowest procyclicality of the entire sample (0.46). The interaction terms are positive, although statistically insignificant, for FS and REF.

6. Comparison with Cincinelli et al. (2021)

Period	Entity	Our Result	Cincinelli (2021)
MPR	CB	0.7833***	0.8244***
MPR	REF	0.6845	0.7466***
PMPR	CB	0.7074***	0.5438***
GFC	CB	0.6396***	1.0535***

The results for the MPR period show excellent alignment with Cincinelli et al. (2021), validating our methodology. There are some differences concerning other periods, which can be attributed mainly to the reduced composition of our sample under analysis (158 vs. 264 companies) and variations in the definition of financial leverage.

7. Economic Interpretation

Period	Economic Context	Expected Pattern	Our Result
GFC	Stimulus via banks	CB highest	CB > FS > REF
MPR	Shadow banking boom	CB very high, REF high	CB > REF > FS
PMPR	Deleveraging campaign	General decline	CB still high
COVID	Bank oversight, digitalization	FS should rise, CB fall	FS > REF > CB

The COVID-19 reversal is particularly noteworthy: commercial banks faced stricter oversight and government support came with conditions, while financial services benefited from the digitalization wave with lighter regulatory constraints.

Conclusions

Our analysis reveals significant heterogeneity in the procyclicality of leverage between entity types and crisis periods: commercial banks are the main drivers of procyclicality, particularly during the MPR period ($\beta = 0.78$); procyclicality varies significantly across crisis periods, with the MPR showing the strongest effect and the GFC the weakest; the COVID-19 period shows a unique pattern in which financial services outperformed commercial banks in terms of procyclicality, reflecting structural changes in the regulatory environment; and mean reversion is consistently present in all specifications, confirming that institutions actively manage target leverage levels. Our results are broadly in line with those of Cincinelli et al. (2021), particularly for the MPR period, validating our methodology, albeit with some differences — notably by extending the analysis to include data from the COVID-19 pandemic.

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